Response to the Referee

Manuscript LR20628MR — "Yukawa screening derivation of the bond-valence rule"

We thank the referee for a careful, constructive report. The perspective of a condensed-matter reader from outside the bond-valence community proved especially valuable, and we agree that it matches the readership of Physical Review Materials.

The referee's paired observations—that the presentation was abrupt, and that the Letter was short enough to absorb more explanation—prompted a more extensive expansion than the specific items below. We have enlarged the presentation throughout the manuscript. The point-by-point responses list the specific changes in detail, but additional context has been added at many points beyond those listed, including a derivation of the Yukawa flux factor at first use, status statements distinguishing postulate from approximation from exact identity at each stage, and mechanistic explanations accompanying the validation figures. The expansion also produced two new results. First, the coordination-resolved line family is now closed by explicit intercept and intersection relations, which give the characteristic softness B^* a geometric meaning as the logarithmic dilation rate of the coordination shell with coordination number. Second, a second-root analysis of the screening inversion yields a contact-distance law against Shannon crystal radii, $|\lambda_-^*| = r_{\mathrm{cr}} + c(z)$ with $c(z) \approx 1.05 + 0.09\,z$ Å, anchoring the fitted screening length in a century of independent crystallographic systematics and identifying a valence-dependent effective oxygen size. Both results are validated in the revised Supplemental Material.

Point-by-point response

1. Construction of the coordination-resolved $B(R_0)$ lines.

> *"...the coordination-resolved $B(R_0)$ lines, which play a central role throughout the manuscript... appear almost without motivation. From the text, it is not immediately clear how these lines are constructed numerically. Are they obtained by fixing R_0 and re-optimizing B, or by some other procedure? ...I think the fitting procedure should be described explicitly in the manuscript."*

The coordination-resolved B–R_0 lines were first introduced in the American Mineralogist paper of Li *et al.* (Ref. [19] of the manuscript), where the underlying bond-valence solver and the per-structure fits are developed in detail, and the original manuscript cited that work intending it to supply this context. We agree that the construction must be stated in the manuscript itself. The fitting procedure is now described explicitly in the manuscript, with the attribution to Li *et al.* made explicit at the point of use, and the referee's direct question is answered directly. Both parameters are fitted jointly by linear least squares for each structure, not by re-optimizing B at fixed R_0 (Empirical validation). Each DFT-relaxed structure thus contributes one (R_0, B) pair, and the coordination-resolved line of a species at coordination n is an outlier-robust regression of B on R_0 across those per-structure pairs (RANSAC consensus refit by least squares on its inliers, at least five structures per cell), with a fitted slope accepted only with at least four inlier structures, slope standard error at most 5, and $R^2 \geq 0.3$. The slopes plotted in Fig. 1 are these regression slopes, and a pointer to this construction now appears at their first use in the manuscript (Parameter-free slope). A new Supplemental Material paragraph ("Line-fitting procedure") documents the complete pipeline, including the coordination binning, the RANSAC residual threshold, the bootstrap error estimation, and the structure-count weights.

2. Broader discussion of previous work.

> *"…the Introduction would benefit from a broader discussion of previous work aimed at providing a physical interpretation of the bond-valence model…so that the reader can more clearly appreciate what is genuinely new in the present contribution."*

The manuscript's introduction has been rewritten. It now states concretely what Preiser *et al.* established with point-charge calculations, characterizes Brown's development of the flux picture into a capacitor-based framework, and distinguishes the Adams line of work as relating B to bond stiffness rather than to screening in the surrounding medium. It also adds the context that makes the novelty legible. The near-degeneracy of jointly fitted (R_0, B) is well documented and has commonly been suppressed by fixing $B \simeq 0.37$ Å, whereas the present work treats that correlation itself as physical content and predicts its slope with no fitted constants. The opening paragraph now also notes the model's growing use as a structural descriptor in machine-learning studies, with three added references. Referencing of previous work has been improved throughout, including the explicit attribution of the B–R_0 line construction to Li *et al.* (Ref. [19]) at its point of use.

3. Exact reformulation vs. approximation vs. interpretation.

> *"…the partition-function formulation and the definition of the first-principles screening centroid are mathematically interesting, but their physical motivation is introduced rather quickly. Readers unfamiliar with bond-valence theory may struggle to distinguish what is an exact reformulation, what is an approximation, and what is a physical interpretation."*

The manuscript now makes this bookkeeping explicit at every stage. Its introduction closes with a roadmap stating that the screened flux-ratio partition is a physical postulate, that the isotropic Yukawa weight and its linearization are the two controlled approximations, and that the partition-thermodynamic relations are exact identities of the bond-valence equation. The partition equation is followed by an exactness statement locating the first mathematical approximation in the linearization. The manuscript's Partition thermodynamics section now opens by motivating the ensemble reading (it supplies the reference length left open in the derivation and the shell radius that the first-principles comparison requires), declares its identities exact with the thermodynamic language as interpretation, and notes where the screened-flux and bond-valence forms coincide only to first order. The screening centroid r_{eff} is now introduced in the manuscript (First-principles confirmation) by its physical role—if bond valence is captured screened flux, the shell radius resolved by the partition should track where screening charge accumulates along each bond—before its operational definition.

4. Spread near $z = 3$ in Fig. 1 and slope uncertainties.

> *"The overall agreement with the predicted slope is encouraging, but there remains a substantial spread for some oxidation states, particularly around $z = 3$…it would therefore be useful to discuss the origin of this scatter more explicitly, and, if possible, to indicate the uncertainty associated with the fitted slopes."*

Both requests are implemented, in the manuscript and in the Supplemental Material, and Fig. 1 itself has been redesigned. The manuscript's Fig. 1 is now a box-and-whisker summary of the per-species slopes at each formal charge, so the spread the referee noted is displayed rather than implied, and the full per-species scatter has moved to the Supplemental Material ("Per-species slopes at fixed coordination") with species-resolved error bars. A new manuscript section, "Origin of the near-pole spread," analyzes the scatter. The pole form itself sets the pattern. A species-level perturbation u of the inverse slope shifts the fitted slope by $-\beta^2 u$, so fixed chemical scatter is amplified quadratically toward the pole, and across the charge classes of Fig. 1 the per-class spread indeed grows as $|\beta|^{1.8}$. The $z = 3$ class at $n = 4$ is the clearest case. The same cations at $n = 6$ match the prediction to a median $|\Delta\beta| = 0.03$, so chemical diversity alone is not the origin. The largest deviations belong to species for which fourfold oxygen coordination is rare (chiefly the trivalent lanthanides and Y, median 17 structures per fit), whereas well-populated trivalent species lie on the curve. Slope uncertainties are now reported in the manuscript (Parameter-free slope) and documented in the Supplemental Material ("Slope standard errors"). Each fitted slope carries a bootstrap standard error, with median relative standard error 0.2% across the 150 slopes and 90th percentile 4%, so the visible scatter reflects real differences among species rather than fit uncertainty.

5. Near-coincidence of the coordination-dependent lines.

> *"...for many species the coordination-dependent $B(R_0)$ lines are almost coincident, making the determination of their intersection appear rather delicate. This may simply reflect the logarithmic dependence of the predicted slopes on z/n... if so I think it would be worthwhile to comment on this point explicitly."*

The referee's surmise is exactly right, and we now comment on it explicitly in the manuscript (Characteristic intersections) and in the Supplemental Material ("Near-coincidence of same-branch lines"). Subtracting two predicted slopes gives $\beta^{(n_2)} - \beta^{(n_1)} = \ln(n_2/n_1)/[\ln(z/n_1)\ln(z/n_2)]$, which is small whenever z is well separated from both coordination numbers, so same-branch lines are expected to appear nearly coincident. The manuscript states this alongside the conditioning statement. The intersection remains constrained exactly to the extent that the data resolve the shell dilation, and the characteristic pair is never computed from a single pair of lines—the construction pools all coordination lines of a species with structure-count weights and reports the residual spread σ_B as its conditioning diagnostic. A new Supplemental Material paragraph quantifies the conditioning. Across the 85 species with three or more lines the median σ_B/B^* is 5.8% and 62 fall below 10%, while the 18 species constrained by only two lines have $\sigma_B = 0$ by construction and are flagged as provisional in the table.

6. Supplemental figure captions.

> *"Note: many figure captions in the Supplementary seem to be incorrect."*

We thank the referee for catching this. A systematic audit of every caption against the rendered figure pages confirmed the problem. The atlas pages order panels alphabetically by element within each block, while the captions had assumed atomic-number ordering, so all eight d-block captions misidentified their contents. Every atlas caption now lists its panels explicitly and correctly, the ordering convention is stated in the atlas introduction, and the remaining caption inaccuracies (color versus line-style encodings, species orderings) have been fixed. Supplemental figures and tables now carry S-prefixed numbering to prevent ambiguity with main-text figure references.

7. Letter format.

> *"I am not sure if the Letter format is the most appropriate for this paper. As now, I find it a bit abrupt, but it is also quite short, so maybe the authors can add enough explanations while keeping it a Letter."*

We followed this suggestion. The explanations described above were added while condensing elsewhere, and the revised manuscript stands at approximately 4,200 word-equivalents, within the Letter limit.

Additional changes

Beyond the items above, the deeper revision the report prompted led to the following changes, each logged in the revision record.

- The manuscript's screening-collapse discussion at $z = n$ was corrected. The original text overstated the constraint (the valence-sum rule fixes only the mean bond valence, not each bond), and the manuscript now distinguishes truly degenerate shells, for which a single structure leaves B unconstrained, from nearly degenerate shells, which are satisfied at finite B, with corpus counts for both. Relatedly, the branch change across the pole is now correctly attributed to the sign reversal of the slope β rather than to the sign of B.
- The tabulated screening length λ^* now covers both branches (signed values for the two anti-screening species), and the second root λ^*_- is tabulated for all species, supporting the new contact-distance analysis.
- A sign census of the per-shell fits, a comparison of B^* with published softness parameterizations, and the slope-error and line-conditioning statistics were added to the Supplemental Material, and all quoted statistics are reproduced by scripts in the project repository.
- Fig. 2 was redrawn for readability (smaller markers, labels adjacent to their points), the offset of the centroid comparison from the identity line is now explained mechanistically with the directly measured oxygen-side shift, and terminology and notation were unified (Gibbs ensemble language, reference length R' distinguished from the characteristic intercept R_0^*).

We believe the revised manuscript is substantially more accessible, and we thank the referee for a report that improved it well beyond the specific points raised.